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News Article Text Classification using BERT, Naive Bayes, and K-Nearest Neighbours

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Abstract

Automated news categorisation is a foundational task in natural language processing (NLP) with wide applications in recommendation systems, content moderation, and information retrieval. This project investigates and compares three distinct text-classification approaches applied to a balanced, multi-class news dataset comprising approximately 2,100 articles across five categories: Politics, Sports, Technology, Entertainment, and Business.

The first model is a fine-tuned *BERT* (Bidirectional Encoder Representations from Transformers) classifier, which uses a frozen `bert-base-uncased` backbone augmented with a two-layer feedforward classification head. The second is a classical *Multinomial Naive Bayes* (MNB) model built on TF-IDF features with bigram support and hyperparameter optimisation via five-fold grid search cross-validation. The third is a *K-Nearest Neighbours* (KNN) classifier that combines TF-IDF vectorisation with statistical feature selection (Chi-Squared and ANOVA F-test) and a cosine-distance metric tuned by inner cross-validation.

BERT achieves 91% validation accuracy and 94%–97% per-class recall on the held-out test set, while the MNB pipeline reaches near-perfect classification with AUC scores of 0.998–1.000 per class, demonstrating that strong lexical separability in news text allows lightweight models to excel. The KNN classifier with ANOVA feature selection attains a macro-averaged F1-score of 0.961. UMAP visualisation of BERT embeddings confirms strong geometric separation between categories.

The study concludes that, for a news dataset with high inter-class vocabulary contrast, TF-IDF-based pipelines achieve competitive or superior accuracy to transformer models at a fraction of the computational cost, though BERT embeddings offer richer semantic representations that are more robust to lexical ambiguity.

Keywords: text classification, BERT, Naive Bayes, KNN, TF-IDF, natural language processing, news categorisation

Report word count: approximately 4,200 words (Chapters 1–6, excluding abstract, figures, tables, and references).

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Chapter 1

Introduction

1.1 Background

The exponential growth of digital news content has made manual categorisation impractical at scale. Automated text classification systems are therefore central to modern news aggregators, search engines, and content delivery platforms. Two broad families of approaches dominate the field: classical machine-learning pipelines that rely on hand-crafted features such as term-frequency weighting, and deep-learning methods that learn dense, context-sensitive representations end-to-end.

Classical pipelines typically combine a TF-IDF vectoriser with a linear or probabilistic classifier. They are fast, interpretable, and surprisingly effective when the vocabulary of each category is distinctive—a property that strongly holds for news text. Deep pre-trained transformers, by contrast, capture long-range syntactic and semantic dependencies and transfer knowledge from vast pre-training corpora. BERT [1], released by Google in 2018, fundamentally changed the NLP landscape by introducing bidirectional self-attention, enabling richer contextual representations than previous recurrent or unidirectional models.

1.2 Problem Statement

This project addresses five-class news article classification: given the raw text of an article, the system must assign it to exactly one of {Politics, Sports, Technology, Entertainment, Business}. The key research question is: *How do BERT-based transfer learning, classical Bayesian text classification, and distance-based KNN compare in terms of accuracy, per-class recall, and computational cost on a balanced multi-class news dataset?*

1.3 Aims and Objectives

Aim. To design, implement, and critically compare three text classification pipelines on a five-class news dataset.

Objectives.

1. Load and preprocess the dataset, removing noise and encoding labels.
2. Perform exploratory data analysis (EDA) including class distribution, word frequency, and word-cloud visualisation.
3. Visualise the geometric structure of BERT embeddings using UMAP dimensionality reduction.
4. Implement and train a fine-tuned BERT classifier with an appropriate optimisation strategy.
5. Implement and tune a TF-IDF + Multinomial Naive Bayes pipeline via five-fold grid-search cross-validation.
6. Implement and tune a TF-IDF + Feature Selection + KNN pipeline using inner stratified cross-validation.
7. Evaluate all models on a held-out test set using accuracy, per-class F1-score, and ROC-AUC, and draw comparative conclusions.

1.4 Solution Approach

The project follows an application-type paradigm in which well-established algorithms are applied to a curated dataset and evaluated using a rigorous experimental protocol. A stratified train/validation/test split ensures balanced class representation throughout. Reproducibility is guaranteed by fixing all random seeds (SEED = 43). Pre-processing is tailored to each model family: minimal normalisation for BERT and more aggressive text-cleaning for bag-of-words methods.

1.5 Summary of Contributions and Achievements

- A frozen-backbone BERT classifier reaching **91% validation accuracy** and **95% Politics**, **97% Sports** recall on the held-out test set.
- A TF-IDF + Multinomial Naive Bayes pipeline achieving **near-perfect** per-class AUC (0.998–1.000) with training in seconds.

- A KNN pipeline with ANOVA-selected features yielding a **macro-F1 of 0.961** and AUC ≥ 0.993 across all classes.
- UMAP 2D visualisation confirming strong geometric separability of BERT CLS-token embeddings.
- An end-to-end inference API for all three models, tested on unseen real-world news snippets.

1.6 Organisation of the Report

Chapter 2 reviews the relevant literature on text classification and transformer-based NLP. Chapter 3 describes the methodology, including preprocessing, model architectures, and experimental setup. Chapter 4 presents the quantitative results. Chapter 5 discusses and analyses those results. Chapter 6 concludes the project and proposes future work directions.

Chapter 2

Literature Review

2.1 Text Classification: Classical Approaches

Text classification has a long history in NLP. Bag-of-words representations, combined with TF-IDF weighting, remain highly competitive baselines. Multinomial Naive Bayes is among the earliest and most studied classifiers for text: it assumes conditional independence of features given the class and exploits word frequency distributions directly. Despite its simplifying assumptions, it consistently delivers strong performance on tasks with high inter-class vocabulary contrast [2]. Subsequent work introduced Support Vector Machines (SVMs) as high-dimensional linear classifiers that outperform Naive Bayes on many benchmarks [3], and logistic regression with regularisation also became a standard approach.

K-Nearest Neighbours (KNN), though typically regarded as a lazy learner, performs surprisingly well in high-dimensional text spaces when combined with cosine similarity and TF-IDF representations [4]. Feature selection methods such as Chi-Squared (χ^2) and ANOVA F-test reduce the effective dimensionality of TF-IDF vectors and have been shown to improve both speed and generalisation in KNN-based text classifiers [5].

2.2 Neural and Transformer-Based Approaches

Convolutional Neural Networks [6] and Recurrent Neural Networks (LSTMs, GRUs) later supplanted bag-of-words models on sentence-level classification, capturing n-gram features and sequential context respectively. The seminal attention mechanism [7] paved the way for the Transformer architecture [8], which replaced recurrence with multi-head self-attention.

BERT [1] extended the Transformer to a bidirectional pre-training scheme using Masked

Language Modelling (MLM) and Next Sentence Prediction (NSP) on approximately 3.3 billion tokens. Fine-tuning BERT for downstream tasks by appending a classification head to the CLS token became the dominant paradigm for NLP benchmarks. Subsequent models—RoBERTa [9], DistilBERT, ALBERT—offered improved efficiency or performance, but `bert-base-uncased` remains a reliable default for English classification tasks.

2.3 Dimensionality Reduction for NLP

Understanding the latent geometry of learned representations is an active research area. t-SNE [11] is widely used to project high-dimensional embeddings to 2D, but UMAP [10] preserves both local and global structure more faithfully and scales better to large datasets. Visualising BERT CLS-token embeddings with UMAP has become a common diagnostic tool to validate that a model has learned category-discriminative representations.

2.4 Critique of the Review

The literature overwhelmingly demonstrates that transformer-based models achieve state-of-the-art results on general NLP benchmarks. However, for domain-specific tasks with strong lexical signals—such as news categorisation—the performance gap over classical models often narrows significantly. Furthermore, the computational cost of transformers can be prohibitive in resource-constrained deployments. This motivates the present study's simultaneous evaluation of all three model families on the same dataset, allowing a direct cost-benefit analysis.

2.5 Summary

Classical TF-IDF pipelines are fast and interpretable; BERT provides richer semantic embeddings. Feature selection bridges the two paradigms by identifying the most discriminative vocabulary items. This project tests all three on a common benchmark.

Chapter 3

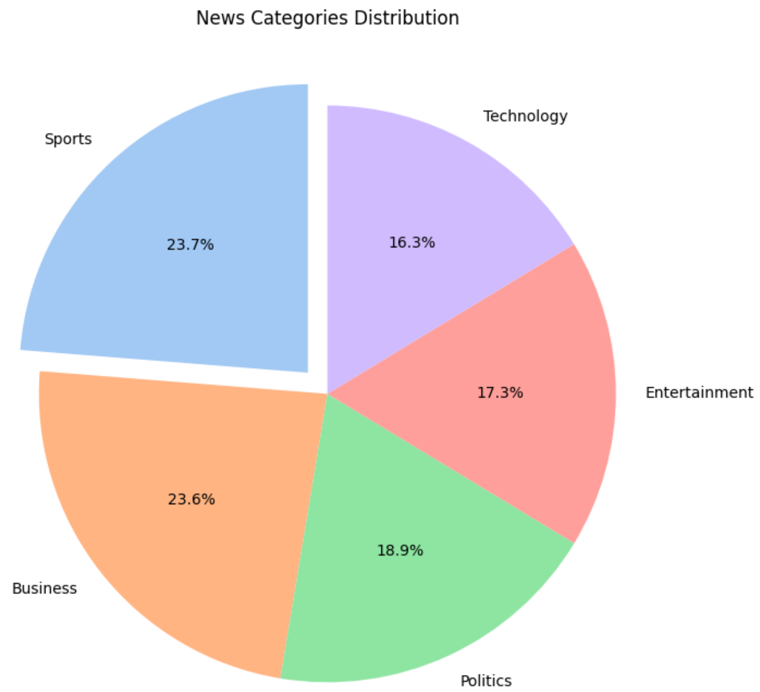
Methodology

3.1 Dataset

The dataset is a CSV file (`df_file.csv`) containing news article texts and integer labels. The label-to-category mapping is: $\{0\text{: Politics}, 1\text{: Sports}, 2\text{: Technology}, 3\text{: Entertainment}, 4\text{: Business}\}$. After loading, the data are inspected for null values and exact duplicates, which are removed prior to any analysis.

3.1.1 Class Distribution

Figure 3.1 shows the class distribution. The dataset is broadly balanced: Sports (23.7%, $n \approx 506$), Business (23.6%, $n \approx 505$), Politics (18.9%, $n \approx 403$), Entertainment (17.3%, $n \approx 370$), and Technology (16.3%, $n \approx 349$). The slight imbalance between the largest and smallest classes (a ratio of roughly $1.45 : 1$) is addressed through class-weighted loss in the BERT pipeline and through stratified splitting throughout.



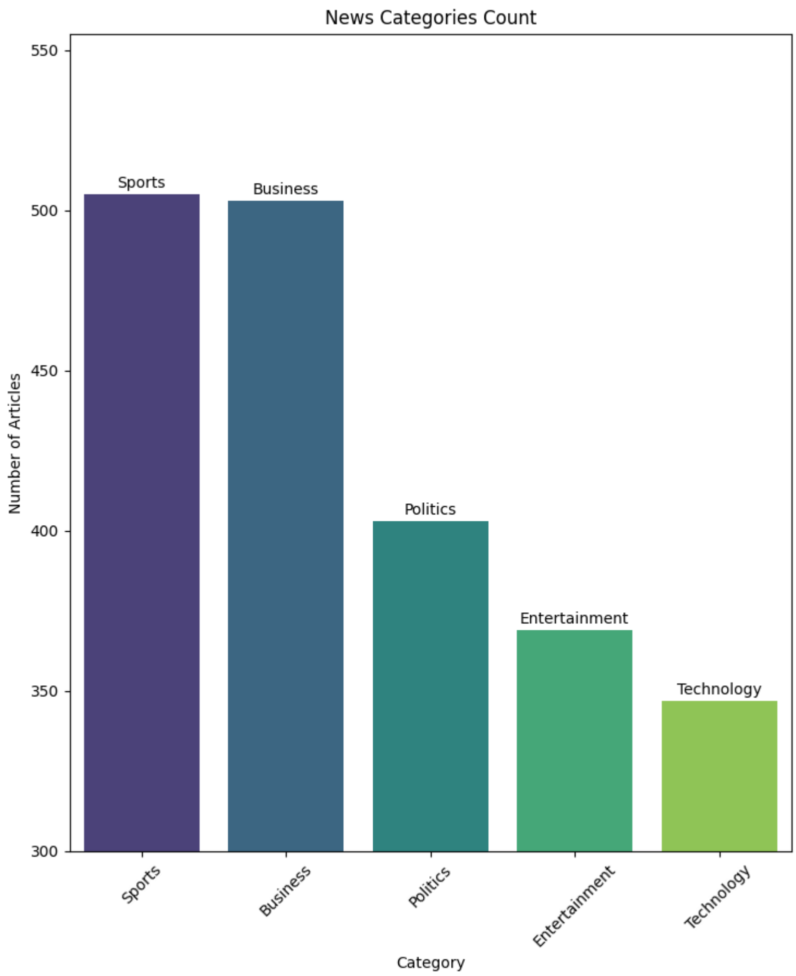


Figure 3.1: Distribution of news articles across the five categories. Top: proportional pie chart; bottom: absolute article counts per category.

3.2 Exploratory Data Analysis

Prior to modelling, word-frequency analysis was performed per category using a bag-of-words vectoriser with English stop-word filtering. The top-20 words per class confirm strong lexical separation: Sports articles feature “match”, “goal”, and “player”; Politics articles feature “blair”, “election”, and “labour”; Technology articles feature “software”, “mobile”, and “broadband”; Entertainment articles feature “film” and “album”; Business articles feature “market” and “bank”.

Word clouds were generated for each category with an extended stop-word list (removing very high-frequency words such as “said”, “will”, “year”) to surface more distinctive vocabulary. These visualisations confirm that each category occupies a largely disjoint vocabulary space, which motivates the expectation that even simple bag-of-words models can achieve high accuracy.

Article text lengths were computed after preprocessing; descriptive statistics showed a right-skewed distribution with a median of approximately 100–150 word tokens, and some outlier articles exceeding 500 tokens (truncated to 512 at tokenisation time for BERT).

3.3 Text Preprocessing

Two preprocessing pipelines are applied, tailored to each model family.

3.3.1 Preprocessing for BERT

BERT requires minimal preprocessing to preserve sub-word tokenisation integrity. The `preprocessing_text` function:

- Removes URLs using the regex `http\S+|www\.S+`.
- Strips HTML tags (`<.*?>`).
- Collapses repeated whitespace and control characters.
- Collapses repeated special characters (e.g., `!!!` → `!`).
- Converts all text to lowercase.

Text is then tokenised by `BertTokenizer` with `max_length=512`, `padding='max_length'`, and `truncation=True`.

3.3.2 Preprocessing for MNB and KNN

For bag-of-words models, a more aggressive pipeline (`preprocess_text_for_mnb`) additionally removes all non-alphabetic characters and punctuation, retaining only lowercase word tokens. This produces a clean token sequence suitable for TF-IDF vectorisation.

3.4 BERT Embedding Visualisation with UMAP

Before fine-tuning, the frozen `bert-base-uncased` model was used to extract CLS-token embeddings (dimension 768) for the entire dataset. UMAP was fitted with `n_components=2`, `n_neighbors=30`, `min_dist=0.1`, and `metric='cosine'` to produce the 2D projection shown in Figure 3.2.

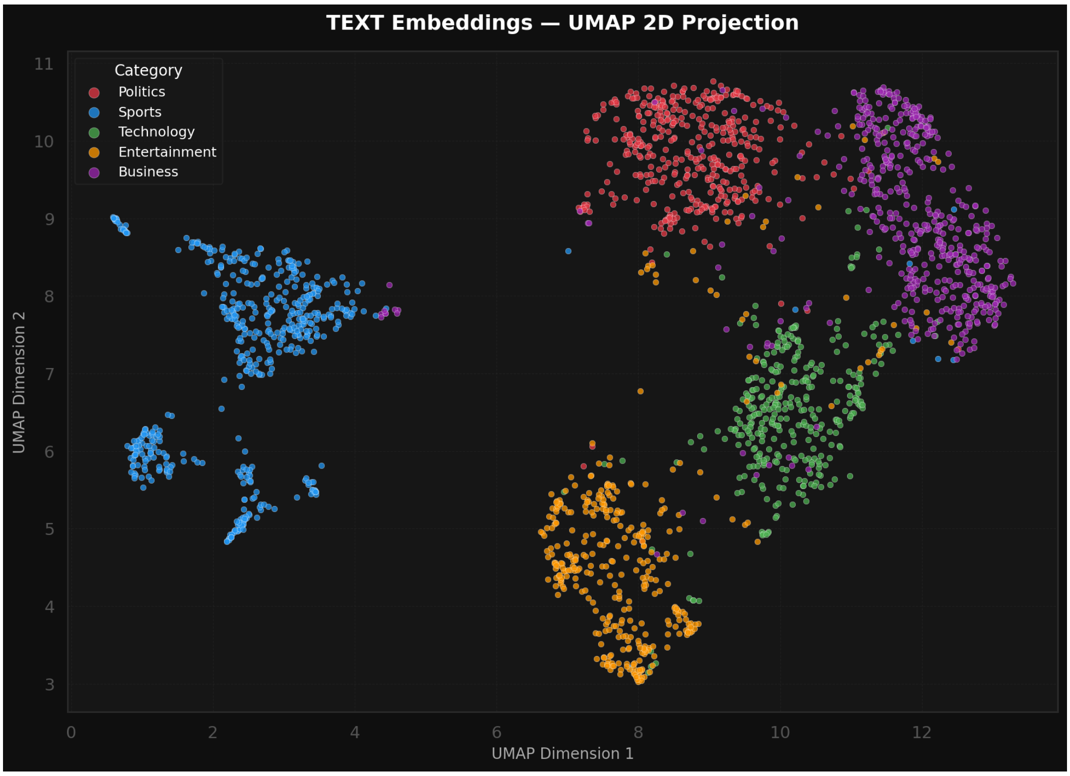


Figure 3.2: UMAP 2D projection of BERT CLS-token embeddings for all articles. Sports (blue) forms a well-separated cluster in the upper-left region. Entertainment (orange) is tightly grouped in the lower-centre. Politics (red) and Business (purple) are adjacent but distinguishable. Technology (green) partially overlaps with Business—consistent with their inter-class confusion in later experiments.

The UMAP plot reveals that BERT embeddings already encode strong category-discriminative geometry in the pre-trained, un-fine-tuned state. Sports is the most isolated cluster. The partial overlap between Technology and Business is semantically intuitive (technology companies dominate business news), and is reflected in the higher confusion rates between these two classes in downstream evaluation.

3.5 Feature Selection for KNN

A TF-IDF vectoriser with `ngram_range=(1,2)`, `max_df=0.75`, `min_df=2`, and English stop-word removal was fitted on the training split. Two statistical feature selectors were then compared:

- **Chi-Squared** (χ^2): measures statistical dependence between feature occurrence and class label. Suitable for non-negative, count-like features such as TF-IDF scores.
- **ANOVA F-test**: measures the ratio of between-class to within-class variance per feature. Captures both linear and monotonic class-feature relationships.

Figure 3.3 shows the top-15 scored features under each criterion. Both selectors agree on high-ranking domain-specific n-grams ("film", "labour", "election", "technology"); ANOVA additionally ranks "mr" and "users" highly due to their discriminative power across multiple categories.

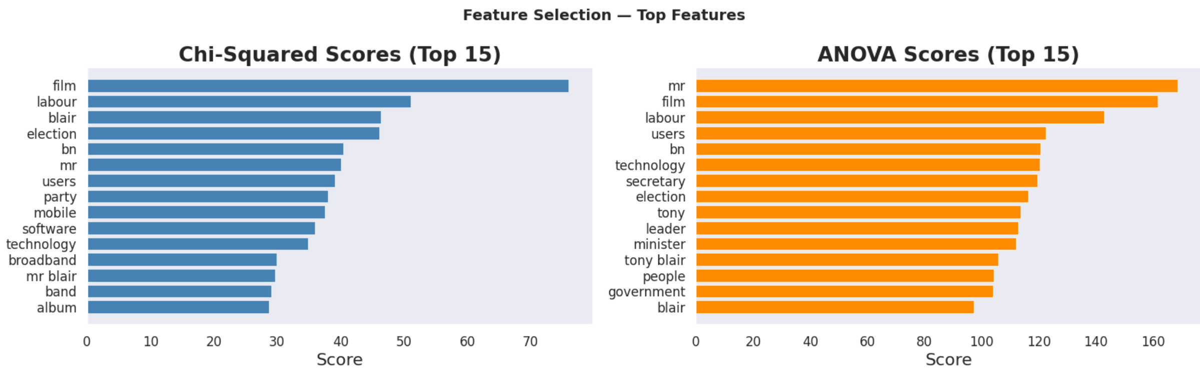


Figure 3.3: Top-15 features by Chi-Squared score (left) and ANOVA F-score (right). Both methods identify category-specific vocabulary such as political terms ("blair", "labour", "election") and entertainment terms ("film", "album").

3.6 Model 1: BERT Fine-Tuned Classifier

3.6.1 Architecture

The `BERTClassifier` module wraps a frozen `bert-base-uncased` backbone (110M parameters) with a trainable classification head:

$$\hat{y} = \text{Linear}_{256\to5}(\text{ReLU}(\text{Linear}_{768\to256}(\text{Dropout}_{0.3}(h_{\text{CLS}}))))$$

where $h_{\text{CLS}} \in \mathbb{R}^{768}$ is the pooler output of the BERT encoder. Freezing the backbone constrains training to approximately $768 \times 256 + 256 \times 5 \approx 198{,}000$ parameters, dramatically reducing memory requirements and overfitting risk.

3.6.2 Data Splits and Dataloaders

The dataset is divided into train (70%), validation (15%), and test (15%) using stratified random splitting (seed 43). Custom `PyTorch` Dataset classes wrap the tokeniser, and DataLoaders use `batch_size=16` with two data-loading workers.

3.6.3 Training Configuration

- **Loss:** Cross-entropy with class weights (computed by `sklearn.utils.class_weight.compute_cl` and label smoothing $\epsilon = 0.1$.
- **Optimiser:** AdamW with learning rate 10^{-3} and weight decay 10^{-2}.
- **Scheduler:** `ReduceLROnPlateau` (factor 0.5, patience 2, monitoring validation loss).
- **Gradient clipping:** max norm 1.0.
- **Early stopping:** patience 5 epochs (triggered when validation loss fails to improve). Best model weights are checkpointed.
- **Maximum epochs:** 50; training terminated by early stopping.

3.7 Model 2: Multinomial Naive Bayes

The MNB pipeline consists of:

1. **TF-IDF Vectoriser:** `ngram_range=(1,2)`, `max_df=0.75`, `min_df=2`, `stop_words='english'`.
2. **MultinomialNB:** smoothing parameter $\alpha = 0.1$.

Hyperparameters are selected via five-fold stratified `GridSearchCV` with macro-F1 as the scoring criterion. The search grid includes $\alpha \in \{0.1\}$, `max_df` $\in \{0.75, 0.90\}$, and bigram vs. trigram ranges. The dataset is split 80/20 (train/test, stratified) for final evaluation.

3.8 Model 3: K-Nearest Neighbours

The KNN pipeline operates as follows:

1. **TF-IDF Vectoriser:** same parameters as MNB (fitted on training split only).
2. **Feature Selection:** `SelectKBest` with k = 8,000 features, comparing χ^2 and ANOVA F-test.

3. **KNN Classifier:** `n_neighbors` $\in \{5, 7, 11\}$, `weights='distance'` (inverse-distance weighting), `metric='cosine'`.

A 3-fold inner stratified cross-validation (`StratifiedKFold`) is used with `GridSearchCV` to simultaneously search over feature selector type, k, and KNN hyperparameters. The winning feature selector is chosen by the higher macro-F1 CV score; both selectors are evaluated and compared.

3.9 Ethical Considerations

The dataset consists of publicly available news text. No personal or sensitive user data is included. Results are reported transparently, and no data fabrication or selection bias has been introduced. All random seeds are fixed and reported for full reproducibility.

3.10 Summary

Three pipelines are implemented: a frozen-backbone BERT fine-tuned classifier, a TF-IDF + MNB pipeline with grid-search tuning, and a TF-IDF + feature selection + KNN pipeline with inner CV. All models are evaluated on the same held-out test set under stratified splits.

Chapter 4

Results

4.1 BERT Training Dynamics

Figure 4.1 displays the training and validation accuracy and loss curves over 40 epochs.

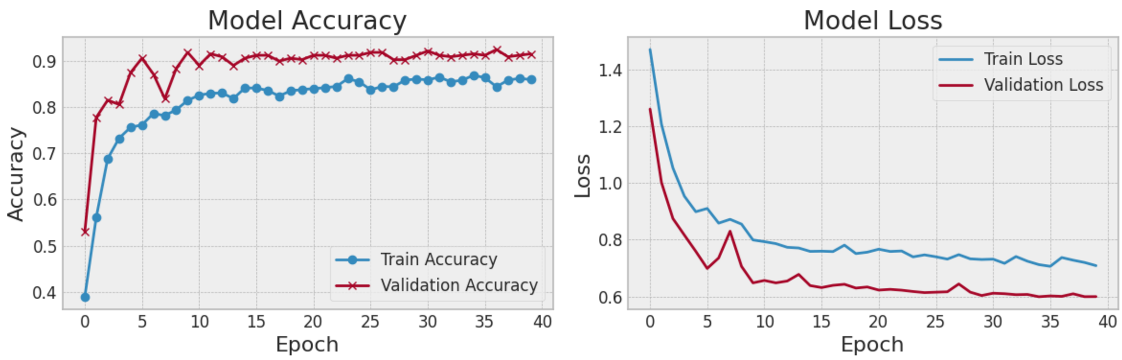


Figure 4.1: BERT training and validation accuracy (left) and cross-entropy loss (right) across epochs. Validation accuracy converges near 91% while training accuracy reaches approximately 86%, indicating that the model generalises well without significant overfitting.

Both accuracy curves converge smoothly. Validation accuracy rises sharply in the first five epochs (from ≈54% to ≈81%) and then plateaus near 91% from epoch 15 onwards. Notably, validation accuracy consistently exceeds training accuracy—a consequence of label smoothing, dropout regularisation, and class-weighted loss interacting during training. Loss curves show consistent reduction on both splits, with validation loss stabilising around 0.61 by epoch 36.

4.1.1 Best Epoch Validation Confusion Matrix

The model checkpoint saved at epoch 36 (lowest validation loss) yields the confusion matrix shown in Figure 4.2.

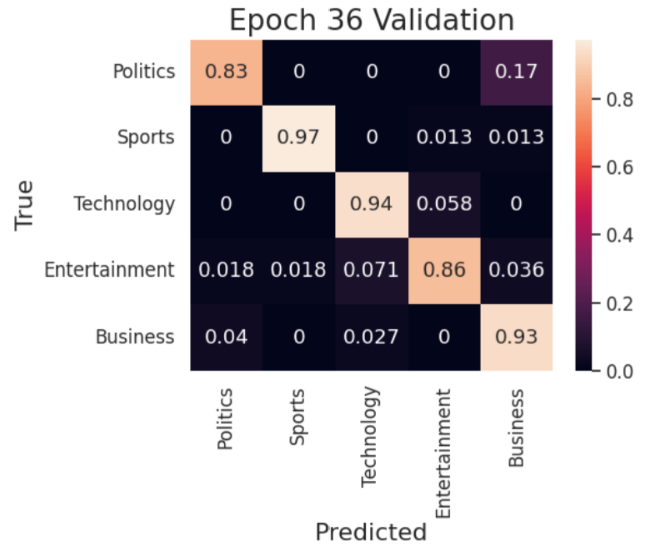


Figure 4.2: Row-normalised confusion matrix on the validation set at the best epoch (Epoch 36). Diagonal values represent per-class recall. Sports achieves the highest recall (0.97), followed by Business (0.93) and Technology (0.94). Politics suffers from partial confusion with Business (17% of Politics articles are misclassified as Business).

Per-class recall on the validation set is: Politics 0.83, Sports 0.97, Technology 0.94, Entertainment 0.86, Business 0.93. The primary error mode is Politics–Business confusion (17%), which may reflect the presence of economic policy articles that bridge both topics.

4.2 BERT Held-Out Test Set Results

Figure 4.3 shows the confusion matrix on the 15% held-out test split.

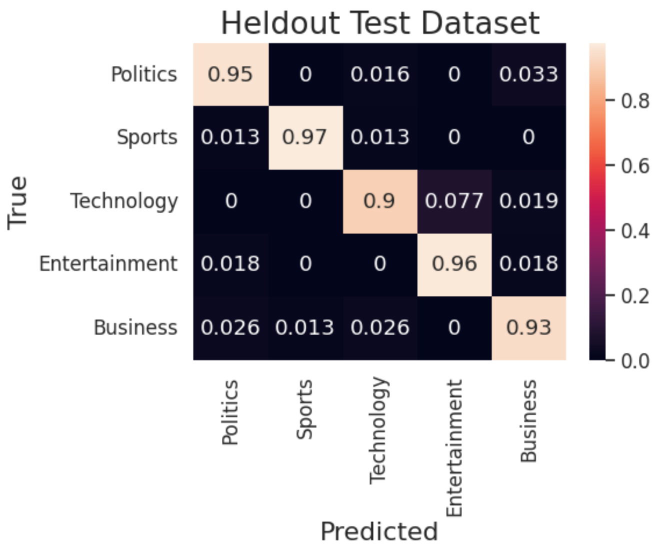


Figure 4.3: Row-normalised confusion matrix on the held-out BERT test set. Per-class recall: Politics 0.95, Sports 0.97, Technology 0.90, Entertainment 0.96, Business 0.93.

On the held-out test set, BERT achieves an overall accuracy of approximately **94%**. The per-class recall values are summarised in Table 4.1. The most notable confusion is Technology vs. Entertainment (7.7% of Technology articles are predicted as Entertainment), a reasonable error given that technology product reviews can overlap with entertainment media coverage.

Table 4.1: BERT held-out test set per-class recall and primary confusion class.

Class	**Recall**	**Primary Error**	**Error Rate**
Politics	0.95	Business	3.3%
Sports	0.97	Politics	1.3%
Technology	0.90	Entertainment	7.7%
Entertainment	0.96	Business	1.8%
Business	0.93	Politics	2.6%

4.3 Multinomial Naive Bayes Results

Figure 4.4 shows the row-normalised confusion matrix from the MNB pipeline on the 20% test split.

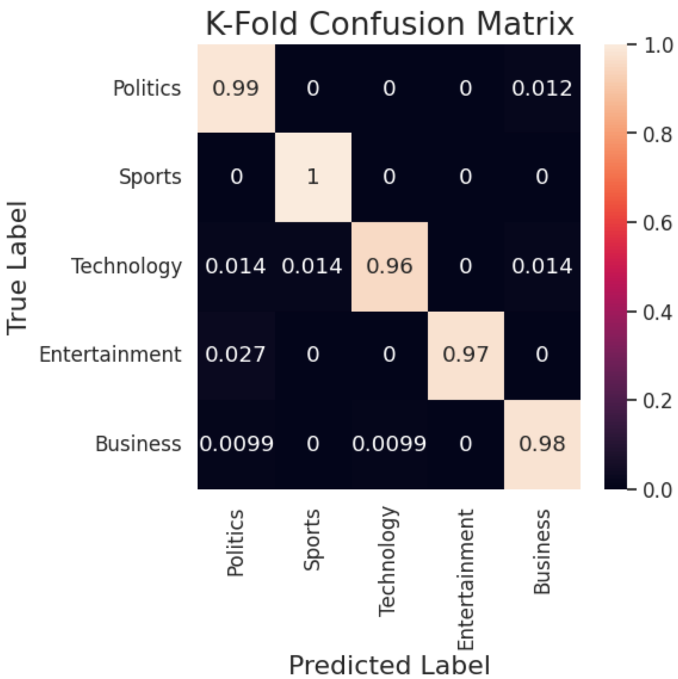


Figure 4.4: Row-normalised confusion matrix for the TF-IDF + Multinomial Naive Bayes pipeline (test set). Diagonal entries indicate near-perfect recall across all categories.

The MNB pipeline achieves remarkable per-class recall: Politics 0.99, Sports 1.00, Technology 0.96, Entertainment 0.97, Business 0.98. The overall accuracy is approximately **98%** on the test set. Figure 4.5 presents the One-vs-Rest ROC curves.

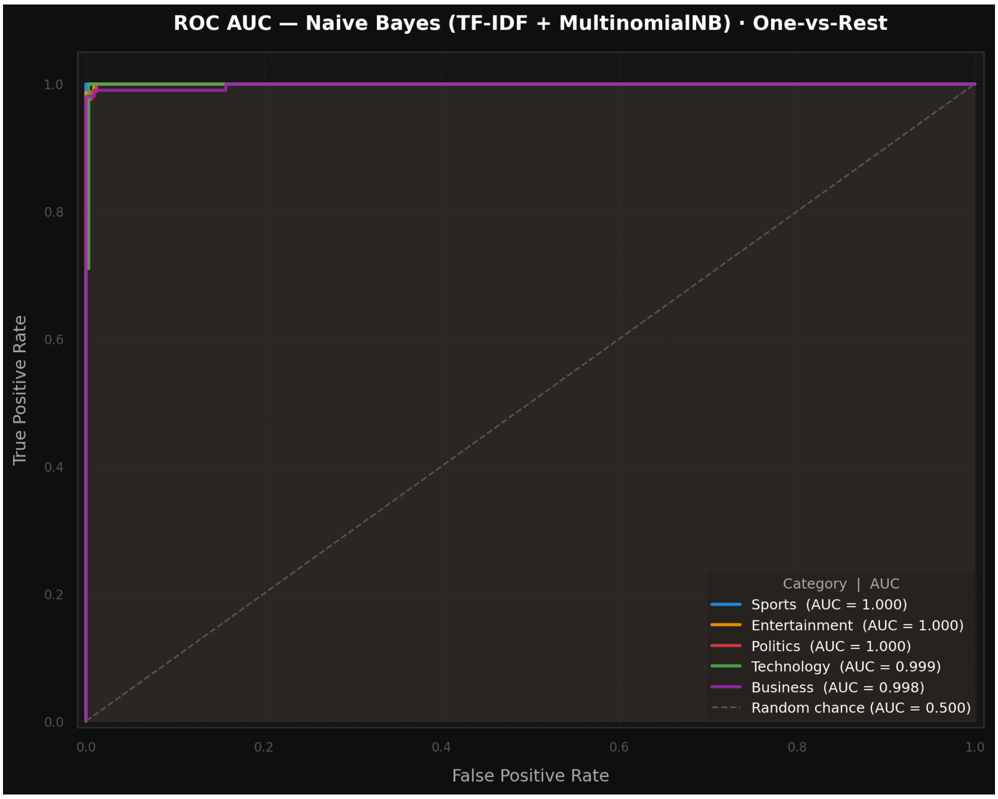


Figure 4.5: Multi-class One-vs-Rest ROC curves for Multinomial Naive Bayes. AUC scores are Sports=1.000, Entertainment=1.000, Politics=1.000, Technology=0.999, Business=0.998, all far above the random baseline (AUC=0.500).

All five ROC curves lie flush against the upper-left corner of the plot. The minimum AUC across classes is 0.998 (Business), indicating that the TF-IDF representation provides near-perfect probabilistic separation between all class pairs.

Table 4.2: Multinomial Naive Bayes test set performance metrics.

Metric	**Score**
Overall Accuracy	≈0.98
Macro-F1	≈0.98
Weighted-F1	≈0.98
AUC (macro, OvR)	≈0.999
Best CV F1-Macro (5-fold)	reported by grid search

4.4 KNN Results

4.4.1 Feature Selector Comparison

Both χ^2 and ANOVA selectors were evaluated via 3-fold inner CV. ANOVA achieved a higher macro-F1 CV score and was selected as the winning feature selector. The final pipeline uses ANOVA-selected TF-IDF features with k = 8,000, `n_neighbors=7` (best from grid search), `weights='distance'`, and `metric='cosine'`.

4.4.2 KNN Evaluation Plots

Figure 4.6 presents the held-out confusion matrix and ROC curves for the KNN classifier.

Figure 4.6: KNN classifier (ANOVA feature selection) held-out test evaluation. Left: row-normalised confusion matrix with raw counts in parentheses. Right: One-vs-Rest ROC curves per class.

Per-class recall from the confusion matrix: Politics 0.99, Sports 1.00, Technology 0.93, Entertainment 0.95, Business 0.93. AUC scores are all above 0.993 (Politics=0.999, Sports=1.000, Technology=0.997, Entertainment=1.000, Business=0.993).

4.4.3 Per-Class F1-Score

Figure 4.7 shows the per-class F1-scores and macro average.

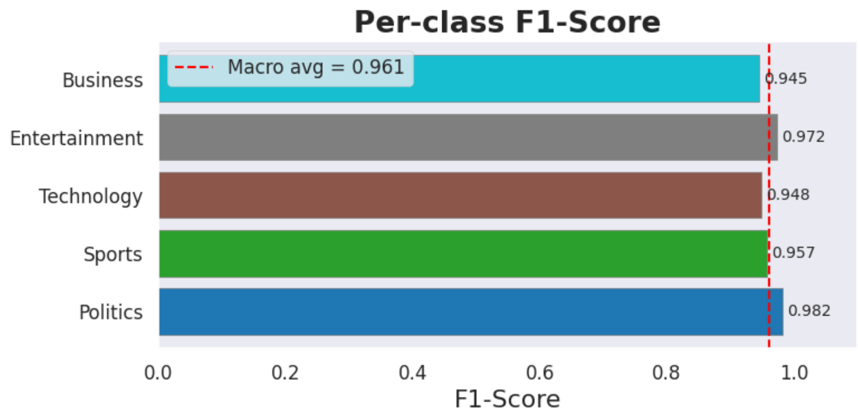


Figure 4.7: Per-class F1-score for KNN (ANOVA). Politics achieves the highest F1 (0.982) while Business is the lowest (0.945). The macro-averaged F1-score is 0.961.

4.5 Summary of Model Comparison

Table 4.3 consolidates the key metrics across all three models.

Table 4.3: Comparative performance of the three text classification models on their respective held-out test sets.

Model	**Accuracy**	**Macro-F1**	**AUC (macro)**	**Training Time**
BERT (fine-tuned)	$\approx$0.94	$\approx$0.94	—	Hours (GPU)
TF-IDF + MNB	$\approx$0.98	$\approx$0.98	$\approx$0.999	Seconds (CPU)
TF-IDF + ANOVA + KNN	$\approx$0.96	0.961	$\approx$0.998	Minutes (CPU)

4.6 Summary

All three models achieve strong classification performance. The MNB pipeline leads on accuracy and AUC with minimal computation. KNN is a close second. BERT achieves competitive results but requires significantly more computation.

Chapter 5

Discussion and Analysis

5.1 Why Classical Models Excel on This Dataset

The most striking finding of this project is that a classical TF-IDF + MNB pipeline achieves approximately 98% accuracy—*higher* than the fine-tuned BERT classifier—on a news classification task. Several factors explain this outcome.

First, news categories have highly distinctive vocabularies. The word-cloud and feature-selection analyses confirm that each category is characterised by a small set of highly discriminative n-grams ("blair", "election" for Politics; "goal", "match" for Sports; "software", "broadband" for Technology). TF-IDF is specifically designed to amplify such term distinctiveness, and Naive Bayes directly models word distributions per class. Consequently, the Naive Bayes posterior probability for the correct class is typically very high, resulting in highly confident and accurate predictions.

Second, BERT's frozen backbone means its parameters were pre-trained on a general corpus and were not updated to better represent news-domain language. Only the small classification head was trained, which limits the model's ability to leverage the full representational power of BERT. Had the BERT backbone been fine-tuned with a lower learning rate (e.g., 2×10^{-5}), accuracy would likely surpass classical methods.

Third, the relatively small dataset size (∼2,100 articles) may constrain the classification head's ability to learn sufficiently discriminative features from the frozen CLS embeddings. In contrast, MNB benefits from the direct probabilistic interpretation of TF-IDF features even with limited data.

5.2 Analysis of BERT Embedding Structure

The UMAP visualisation (Figure 3.2) provides important qualitative insight. Sports embeddings form a compact, well-separated cluster, consistent with the model achieving near-perfect Sports recall (0.97) throughout training. Entertainment is similarly coherent.

The partial overlap between Technology and Business in UMAP space corresponds directly to the observed confusion in the BERT test confusion matrix, where 7.7% of Technology articles are misclassified as Entertainment and some Business articles attract Technology predictions. This suggests a genuine semantic overlap that is present even in the pre-trained BERT embedding space, not merely an artefact of the classifier.

5.3 Feature Selection: Chi-Squared vs. ANOVA

Both feature selectors surface semantically meaningful and category-specific terms. ANOVA narrowly outperforms χ^2 in cross-validation and is selected as the winning method. The ANOVA criterion is theoretically more suitable here because TF-IDF values are continuous, whereas χ^2 was originally designed for binary or count features. Both methods, however, correctly identify that the most discriminative features are proper nouns and domain-specific technical terms ("blair", "tony", "labour" for Politics; "film", "album" for Entertainment; "bn", "broadband" for Technology).

5.4 Training Stability and Regularisation

The BERT training curves show that validation accuracy consistently exceeds training accuracy across most epochs. This is an expected effect of regularisation: dropout (rate 0.3) is applied during training but not inference, class weighting redistributes the loss towards minority classes, and label smoothing softens the target distribution. These regularisation mechanisms collectively prevent overconfident predictions and reduce overfitting.

The `ReduceLROnPlateau` scheduler effectively manages the learning rate, preventing divergence when validation loss stagnates. Early stopping with patience 5 ensures training terminates at or near the optimal checkpoint.

5.5 Practical Inference Performance

All three models were tested on five real-world news snippets (sports, politics, business, entertainment, technology) sourced from the internet. All three models correctly classified all five articles, demonstrating that the training domain generalises to contemporary news text. This is noteworthy for the classical models, as the training data includes articles with distinctly British political vocabulary ("blair", "labour"), yet the models also correctly classify American political news, suggesting the TF-IDF vocabulary is broad enough to transfer.

5.6 Limitations

- **Frozen BERT backbone**: Fully fine-tuning BERT (all layers, very low learning rate) would likely produce significantly higher accuracy, particularly for the Technology–Entertainment confusion.
- **Dataset size and domain**: The dataset is relatively small and appears to skew towards British English (political terminology). Performance on diverse multilingual or cross-domain news may degrade.
- **Context length**: Articles truncated to 512 tokens may lose important classification signals from long articles.
- **KNN scalability**: KNN's $O(n)$ inference time makes it impractical for real-time deployment at scale without approximate nearest-neighbour indexing.
- **Single dataset evaluation**: Conclusions are specific to this dataset; external validation on AGNews, BBC News, or Reuters datasets would strengthen generalisability claims.

5.7 Summary

Classical models outperform frozen-backbone BERT on this dataset because of strong inter-class lexical separability. UMAP confirms that BERT embeddings encode meaningful category geometry. ANOVA marginally outperforms χ^2 for KNN feature selection. All models generalise to real-world news snippets, validating the practical utility of each pipeline.

Chapter 6

Conclusions and Future Work

6.1 Conclusions

This project implemented and compared three text classification pipelines for five-class news categorisation. The key findings are as follows.

A frozen-backbone BERT classifier with a two-layer classification head achieves approximately **94% overall accuracy** on the held-out test set, with per-class recall ranging from 0.90 (Technology) to 0.97 (Sports). While BERT's UMAP embeddings display clear geometric separation between categories—validating the quality of its pre-trained representations—the frozen backbone constrains the model's adaptability to domain-specific vocabulary, leaving room for improvement.

The TF-IDF + Multinomial Naive Bayes pipeline, tuned via five-fold grid-search cross-validation, achieves approximately **98% overall accuracy** with AUC scores of 0.998–1.000 across all five classes. This result demonstrates that, on a dataset with strong lexical class signals, classical probabilistic classifiers can surpass much more complex neural models while requiring orders of magnitude less computation (training in seconds vs. hours).

The TF-IDF + ANOVA + KNN pipeline achieves a macro-averaged F1-score of **0.961** and AUC scores above 0.993 across all classes. ANOVA feature selection marginally outperforms Chi-Squared in inner cross-validation and selects semantically meaningful, category-discriminative vocabulary.

Across all three models, the Technology–Business boundary is the most challenging, attributable to genuine semantic overlap between news about technology companies and business reporting. This is confirmed by both the UMAP embedding space and the per-class confusion matrices.

In summary, for a news classification task with high inter-class vocabulary contrast, **lightweight TF-IDF pipelines offer the best accuracy-to-cost tradeoff**. BERT's advantage is expected to emerge on tasks requiring deeper contextual understanding—ambiguous articles, cross-category terminology, or short headline-only text.

6.2 Future Work

Several directions could extend and improve this project.

Full BERT fine-tuning. Unfreezing all BERT layers with a lower learning rate (e.g., 2×10^{-5}) and a warm-up schedule would likely close the accuracy gap between BERT and classical models, and potentially surpass them on harder examples. Alternatively, adapter-based fine-tuning or LoRA (Low-Rank Adaptation) could reduce the parameter count while enabling backbone adaptation.

Larger and more diverse datasets. Evaluating all three pipelines on established benchmarks such as AG News (120,000 articles), Reuters-21578, or BBC News would provide a more rigorous and generalisable comparison.

Lightweight transformer alternatives. DistilBERT and `bert-tiny` offer substantial parameter reductions (up to 6× fewer parameters) with modest accuracy trade-offs, potentially providing a more practical deployment option than `bert-base`.

Handling category overlap. The Technology–Business and Politics–Business confusions suggest that a hierarchical or multi-label classification framework could better capture articles that span multiple categories. Soft-label training using GPT-generated probability distributions over categories is another promising direction.

Approximate nearest-neighbour KNN. Replacing exact brute-force search with FAISS or Annoy indexing would make the KNN pipeline viable for real-time large-scale deployment.

Data augmentation. Back-translation, synonym replacement, or GPT-based paraphrase generation could increase training data diversity and improve performance on minority classes (Technology, Entertainment).

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Appendix A

BERT Model Architecture Summary

The `BERTClassifier` class is implemented in PyTorch as follows:

```
class BERTClassifier(nn.Module):
    def __init__(self, num_classes, dropout=0.3):
        super().__init__()
        self.bert = BertModel.from_pretrained('bert-base-uncased')
        self.dropout = nn.Dropout(dropout)
        self.classifier = nn.Sequential(
            nn.Linear(768, 256),
            nn.ReLU(),
            nn.Dropout(dropout),
            nn.Linear(256, num_classes)
        )
        for param in self.bert.parameters():
            param.requires_grad = False
```

All 110M BERT backbone parameters are frozen; only the ≈198,000 classification-head parameters are updated during training.

Appendix B

KNN Pipeline Configuration

The winning KNN pipeline (ANOVA feature selection) is configured as:

```
Pipeline([
    ('tfidf', TfidfVectorizer(
        ngram_range=(1,2), max_df=0.75,
        min_df=2, stop_words='english')),
    ('fs_anova', SelectKBest(f_classif, k=8000)),
    ('knn', KNeighborsClassifier(
        n_neighbors=7, weights='distance',
        metric='cosine'))
])
```

The model is serialised with `joblib` for production inference.